Department of Economics

MSc Extended Essay Cover Sheet

CANDIDATE NUMBER *(This is a **5 digit number** found on LFY. It is **not** your 9 digit student number)*	**2**	**6**	**5**	**3**	**7**

COURSE CODE	**E**	**C**	**4**	**2**	**3**

TITLE	**Recreational Marijuana Legalization and Mental Health (Independent Research)**

WORD COUNT	**4**	**4**	**0**	**0**

Double spaced, typed. Maximum 6,000 words (as a guideline this is 14-21 double spaced pages). Abstract, footnotes, references and appendices do not count toward the word count, provided such additions are brief and do not contain information that rightly belongs in the body of the essay. Equations are included in the word count and counted as the page equivalent (i.e., as the number of words that would occupy the same amount of space in text). Examiners are recommended to penalise excessively long essays (+ 10% of word count) by adjusting the marks by 6000/x, where x is the word count. Pages are to be numbered.*

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Recreational Marijuana Legalization and Mental Health

26537[1]

May 31, 2022

Abstract

I take advantage of the variation in recreational marijuana legalization across US states and years to identify its effect on mental health outcomes using a staggered difference-in-differences approach. Using state-level panel data from the National Survey on Drug Use and Health (NSDUH), I find evidence that recreational marijuana laws (RMLs) increase the prevalence of mental illness, major depressive episodes, and suicidal ideation among adults ages 18 and older. These findings are robust across a variety of different specifications and contradict past research linking medical marijuana laws (MMLs) with improvements in mental health, suggesting that RMLs affect the population differently from MMLs. Further research is necessary to understand the mechanisms behind these results and identify groups that are particularly at risk.

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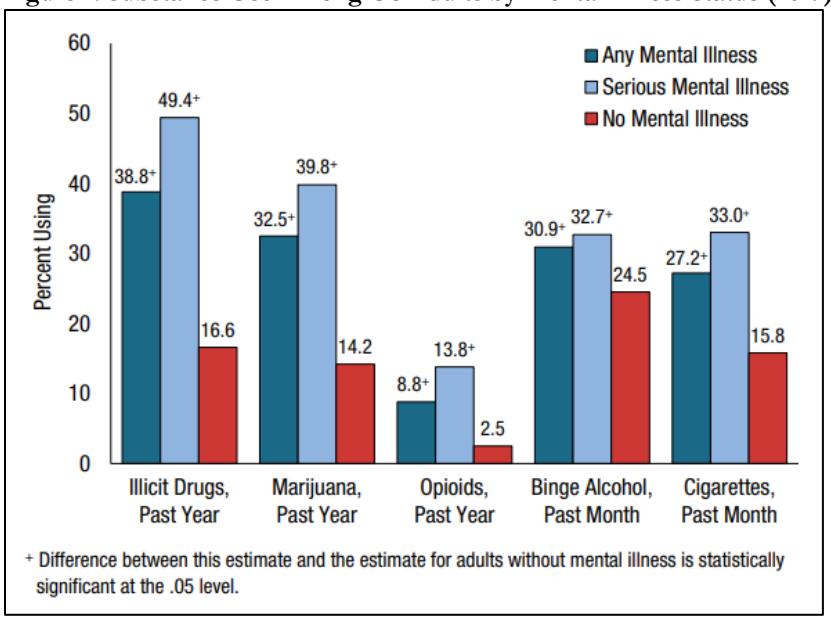
1. Introduction

Epidemiologists have long observed the comorbidity between mental health disorders and substance use and abuse. In 2019, 38.8% of adults with any mental illness and 49.4% of adults with a serious mental illness reported using illicit drugs in the past year, compared to only 16.6% of adults with no reported mental illness (SAMHSA, 2020, p. 47). This pattern extends to marijuana, and evidence exists for positive statistical associations between marijuana use and bipolar disorder, depressive disorders, suicidal ideation, anxiety disorders, schizophrenia, and other psychoses (National Academies of Sciences, Engineering, & Medicine, 2017, pp. 289–326).

Figure 1. Substance Use Among US Adults by Mental Illness Status (2019)



Source: SAMHSA (2020).

However, despite substantial research establishing these associations, much less work has been done determining the direction of causality, which could be driven by a few different, possibly co-occurring explanations:

(1) Marijuana use may increase the risk of developing a mental health disorder, perhaps due to neurobiological changes stemming from use

(2) Mental illness may increase the likelihood of marijuana use; for instance, individuals may use marijuana as a form of self-medication for a mental health disorder

(3) Unobserved genetic and/or environmental factors may predispose certain people to marijuana use as well as to developing mental health disorders (National Academies of Sciences, Engineering, & Medicine, 2017, p. 296)

Studies looking at the relationship between medical marijuana laws (MMLs) and mental health find that MMLs led to improvements or had no effect on mental health outcomes, which would suggest that associations between marijuana use and mental health are mostly driven by explanations (2) and (3). Anderson, Rees, & Sabia (2014) analyze data from the National Vital Statistics System for 1990–2007 and find that medical marijuana legalization was associated with reductions in suicides, particularly for men ages 20–39, of around 9–11 percent.[2] Bartos, Kubrin, Newark, & McCleary (2020) look at the effect of California's MML on suicides using a synthetic control design, finding that MML adoption led to an 11 percent decline in suicides relative to its synthetic control. Using data from the Behavioral Risk Factor Surveilance System, Kalbfuß, Odermatt, & Stutzer (2018) find that MML adoption reduced the number of poor mental health days by 5 percent. Finally Leung (2019) analyzes MML adoption on the mental health of US college students using data from the Healthy Mind Survey, but finds no evidence that MMLs had an effect on the mental health outcomes in the survey.

However, MMLs are an imperfect tool for exploring the relationship between marijuana use and mental health due to the targeted nature of those laws. Since MMLs expand access mainly to the small subset of the population with medical conditions believed to be treatable using marijuana, the external validity of those results may be limited and the size of the effects may be small and difficult to detect. I seek to circumvent those issues by focusing on recreational marijuana laws (RMLs), which remove most restrictions on marijuana consumption and expand marijuana access to the general adult population. By taking advantage of differences in timing among RML adoption across states, I can estimate the effect of RMLs on mental health using the two-way fixed effects model commonly used for estimation within a staggered difference-in-differences setting. I additionally advance on the previous literature through the use of measures of mental health from the National Survey of Drug Use and Health (NSDUH).

I first estimate the effect of RML adoption on marijuana use as a first stage, using state-level data from the NSDUH Small Area Estimate (SAE) files. I find that RMLs led to a 17–20 percent increase in

[2] These results are reexamined by Grucza et al. (2015) using expanded data through 2010, as opposed to 1990–2007 in the original study. They find that the negative association between MML adoption and suicides are no longer statistically significant after controlling for additional covariates, such as race and ethnicity.

reported marijuana use in the past month and a 17–19 percent increase in reported marijuana use in the past year. I then look at the effect of RML adoption on mental health outcomes, and estimate that RMLs are associated with a 4–5 percent increase in the share reporting any mental illness in the past year, a 5–8 percent increase in the share reporting a major depressive episode in the past year, and a 5–9 percent increase in the share reporting serious thoughts of suicide in the past year among adults ages 18 and older. Evidence that RML adoption led to increases in serious mental illness in the past year are less precise and lose statistical significance once I restrict the sample to MML-adopting states and add additional covariates. I also find no evidence that RMLs impact the share of people receiving mental health services. These results appear to contradict the existing literature on MMLs and mental health, and suggests there may in fact be a causal link between marijuana use and declines in mental health.

The remainder of the paper is divided into four sections: In Section 2, I go into detail about the data used in my analysis. In Section 3, I describe my econometric model and identification strategy. In Section 4, I present the results of my analysis. And in Section 5, I conclude with a discussion of the implications of my results along with some of the limitations of my analysis.

2. Data

My measures of mental health come from the National Survey on Drug Use and Health (NSDUH), an annual survey administered by the Substance Abuse and Mental Health Services Administration (SAMHSA), an agency within the US Department of Health and Human Services. The NSDUH interviews around 70,000 people ages 12 and older each year and has covered all 50 states and the District of Columbia since 1999. It is considered the "leading source of statistical information on the use of illicit drugs, alcohol, and tobacco and mental health issues in the United States," and is widely used by government agencies and independent researchers for tracking trends in substance abuse, mental illness, and mental health treatment (SAMHSA, 2020).

While SAMHSA makes NSDUH data available to the public, the public-use data omits any geographic identifiers including state of residence due to confidentiality rules, which precludes me from using this data for my study.[3] Researchers can apply to use the restricted data, which includes geographic identifiers as well as other sensitive variables and additional observations. However, the restricted data can only be accessed from inside a "research data center" within the United States and gaining entry requires an extensive application process, a security clearance, and nontrivial application fees based on the number of years of data requested.[4]

With the requisite individual-level data being prohibitively difficult to access, I instead rely on variables from the NSDUH Small Area Estimate (SAE) dataset. The SAE data contains two-year estimates of prevalence rates for selected substance use and mental health outcomes, aggregated at the state level using results from the NSDUH restricted-use data. While the SAE dataset spans 1999–2000 to 2019–2020, the majority of my measures of interest for mental health are only available beginning in 2008–2009, with the exception of "received mental health services in the past year," which becomes available beginning in 2010–2011. I additionally elect to omit 2019–2020 data from my regressions to avoid any confounders and issues with data quality stemming from to the COVID-19 pandemic.

I use the following measures of marijuana use and mental health from the NSDUH SAE files (SAMHSA, 2020):

[3] https://www.datafiles.samhsa.gov/get-help/public-vs-restricted-use/what-difference-between-nsduh-public-and-restricted-use-data
[4] https://www.samhsa.gov/data/data-we-collect/samhsa-rdc

- "Marijuana use in the past month" measures the share of the population reporting using marijuana (excluding blunts) in the past month
- "Marijuana use in the past year" measures the share of the population reporting using marijuana (excluding blunts) in the past year
- "Any mental illness in the past year" measures the share of the population with a "diagnosable mental, behavioral, or emotional disorder… (excluding developmental and substance use disorders) … regardless of the level of impairment in carrying out major life activities"
- "Serious mental illness in the past year" refers to the subset of the previous group of people whose disorder causes "substantial impairment in carrying out major life activities."
- "Major depressive episode in the past year" measures the share of the population that experienced a period of 2 weeks or longer during which they "felt depressed or lost interest or pleasure in daily activities" alongside other symptoms associated with major depressive disorder
- "Serious thoughts of suicide in the past year" measures the share of the population reporting to have seriously thought about killing themselves
- "Received mental health services in the past year" measures the share of the population reporting "receipt of treatment or counseling for any problem with emotions, nerves, or mental health… or the use of prescription medication to treat a mental or emotional condition"

NSDUH carries many advantages as a source of mental health data. In particular, the detailed information it collects on symptoms can be used to classify mental health disorders based on the same diagnostic criteria used by mental health professionals. NSDUH data does however possess many of the limitations that comes with survey data. For example, since data on drug use and other measures are self-reported, there is the possibility of errors due to over or underreporting. Additionally, the survey excludes active-duty service members and individuals living in institutional group quarters, like hospitals, prisons, nursing homes, and treatment centers who altogether make up approximately 3 percent of the US population.

I use the effective dates for medical and recreational marijuana laws put together by Anderson & Rees (2021), which is current through early-2021.[5] There is some debate about how to best model differences in marijuana legislation across states, particularly MMLs (Anderson & Rees, 2021, pp. 5-9). Given that MMLs are not my primary policy of interest, I elect not to add any additional controls in my main specifications for different characteristics of MMLs across states. Though I do run variations of my models including a control for whether a state had operating, legally protected medical dispensaries as a robustness check and find no substantive changes to my results.[6]

I present descriptive statistics for each of the outcomes and measures used in my regressions in Table 1. I break them out by states which adopted an RML by 2019 in the first column, states which only adopted an MML by 2019 in the second column, and states which had not adopted an RML or MML by 2019 in the third column. In general, we see that RML-adopting states had higher rates of marijuana use, slightly higher rates of any mental illness, and similar rates of serious mental illness, major depressive episodes, serious thoughts of suicide, and mental health service utilization as the MML- and non-adopting states. We also see that by most measures of demographic and state-specific characteristics, RML-adopting states better resemble MML-only states than non-adopting states. Based on this and other likely sources of similarity between RML-adopting and MML-only states, I elect to include specifications restricting the sample of states to MML adopters only.

Table 1. Descriptive Statistics for Data Used in Analyses (2008-2009 to 2018-2019)

	RML-Adopting States	MML-Only States	Non-Adopting States
% Reporting Marijuana Use in the Past Month	11.1%	8.0%	6.6%
% Reporting Marijuana Use in the Past Year	16.9%	12.8%	11.0%
% Reporting Any Mental Illness in the Past Year	18.9%	18.2%	18.5%
% Reporting Serious Mental Illness in the Past Year	4.2%	4.1%	4.3%
% Reporting Major Depressive Episode in the Past Year	6.9%	6.7%	6.9%
% Reporting Serious Thoughts of Suicide in the Past Year	4.1%	4.0%	4.0%
% Receiving Mental Health Services in the Past Year	14.4%	14.6%	14.6%
% Female	50.4%	51.0%	50.8%
% White	70.4%	73.7%	75.4%
% Ages 18-64	63.3%	62.1%	62.0%
% Ages 65+	13.6%	15.6%	13.5%
% College Graduate (4-Year)	32.4%	30.5%	27.8%
% With Health Insurance	88.6%	88.7%	85.6%
Unemployment Rate	7.2%	6.4%	6.0%
Per-Capita Income	49,845	48,138	42,542
Beer Tax	0.20	0.23	0.39
Population	7,018,383	6,120,538	5,799,947
Number of States	11	21	19

Notes: Population weighted. Prevalence rates for marijuana use and mental health measures are for adults aged 18+.

[5] I list the effective dates for state MMLs and RMLs in Appendix A.

[6] This data comes from the RAND-USC Schaeffer Opioid Policy Tools and Information Center (2020).

3. Empirical Strategy

My main specification is the two-way fixed effects model commonly used for staggered difference-in-differences estimation in a panel setting:

$$\ln y_{st} = ML'_{st}\beta + X'_{st}\Gamma + \mu_s + \lambda_t + \theta_{r(s)} \times t + \varepsilon_{st} \tag{1}$$

ML'_{st} is a vector of marijuana laws in state s and at period t. In my models, I include one variable representing the fraction of the period in which an RML was in effect and occasionally a second variable representing the fraction of the period in which an MML was in effect. For instance, if a recreational marijuana law took effect for a state on January 31, 2014, the value of the RML variable for that state in 2014-2015 would be $(365 \times 2 - 31)/(365 \times 2) = 0.958$, and 1 in each subsequent period.[7] X'_{st} is a vector of demographic covariates in state s and at period t, which I include in some of my regressions. For those specifications I include controls for gender, race, and age distribution, educational attainment, unemployment rate, per capita income, alcohol tax rate, and share of population with health insurance coverage.

All my regressions include state and time fixed effects, represented respectively by μ_s and λ_t, while some specifications also contain $\theta_{r(s)} \times t$, representing region-specific time trends for census region r of state s.[8] Finally, y_{st} represents the various dependent variables of interest in state s and at period t. I use log-level regressions, so we interpret the estimated effect of an RML or MML as $\%\Delta y = \left(e^{\beta} - 1\right) \times 100$. I also employ population weighting for all the regressions presented, though I also ran unweighted regressions for each of my models and receive similar estimates.

I additionally employ complementary event studies to estimate lagged effects and test for parallel trends. These take the model form of Equation (1), but extended to include a set of lags and leads in place of the treatment variable:

$$\ln y_{st} = \alpha + \sum_{j=2}^{J} \beta_j (\text{Lag } j)_{st} + \sum_{k=1}^{K} \gamma_k (\text{Lead } k)_{st} + MML'_{st}\delta + X'_{st}\Gamma + \mu_s + \lambda_t + \theta_{r(s)} \times t + \varepsilon_{st} \tag{2}$$

[7] I run variations of my regressions using binary indicator variables for MML and RML and find that they do not produce substantively different estimates.
[8] For a map of US states and census region, see https://www2.census.gov/geo/pdfs/maps-data/maps/reference/us_regdiv.pdf.

Where lags and leads are defined per Clarke & Schythe (2020) as follows:

$$(\text{Lag } J)_{st} = \mathbb{1}[t \leq Event_s - J] \tag{3}$$

$$(\text{Lag } j)_{st} = \mathbb{1}[t = Event_s - j] \text{ for } j \in \{1, \dots, J-1\} \tag{4}$$

$$(\text{Lead } k)_{st} = \mathbb{1}[t = Event_s + k] \text{ for } j \in \{1, \dots, K-1\} \tag{5}$$

$$(\text{Lead } K)_{st} = \mathbb{1}[t \geq Event_s + K] \tag{6}$$

For my specifications I set $J = 2$ and $K = 4$ for lags and leads beyond those points to "accumulate". This prevents the estimation of leads or lags at the extremes using too few observations. I present plots showing the results of these estimates in Section 4.3.

4. Results

4.1 Recreational Marijuana Legalization and Marijuana Use

I first conduct estimations of the effect of RML adoption on marijuana use. These serve as a first stage which will help with interpretation of my "reduced form" results and give additional credence to the validity of my model. I present results of these estimations in Table 2, showing that RMLs are associated with a 0.155–0.185 increase in log past month marijuana use and a 0.153–0.172 increase in log past year marijuana use, equivalent to a 17–20 percent increase in past month use and a 17–19 percent increase in past year use respectively. These results are all significant at 1% level, and consistent with recent research finding increases of marijuana use from RMLs on the order of 25% (Hollingsworth, Wing, & Bradford, 2020). The differences in magnitudes of our estimates most likely reflects idiosyncratic distinctions in model specification.

Table 2. Log Proportion Reporting Marijuana Use in the Past Month and Year, Ages 18+

	Past Month Use		Past Year Use	
	All States	MML Adopters	All States	MML Adopters
	(1)	(2)	(3)	(4)
RML in Effect	0.185***	0.155***	0.172***	0.153***
	(0.0339)	(0.0384)	(0.0295)	(0.0306)
MML in Effect	0.0372	-0.0335	0.0164	-0.0104
	(0.0236)	(0.0390)	(0.0194)	(0.0278)
State & Time FEs	Yes	Yes	Yes	Yes
State-Level Covariates	Yes	Yes	Yes	Yes
Region-Specific Time Trends	Yes	Yes	Yes	Yes
Sample Mean	.0826	.0916	.1318	.1436
Observations	306	192	306	192
R-Squared	0.890	0.927	0.911	0.938

Notes: * p < 0.10, ** p < 0.05, *** p < 0.01.
All regressions include state and year fixed effects and are weighted by state population. Time-varying state-specific covariates include share of the population that is female, share that is white, share aged 18-64, share aged 65+, share with a college degree, unemployment rate, per capita income, alcohol tax rate, and share with health insurance. Standard errors, allowing for clustering at the state level, are reported in parentheses.

4.2 Recreational Marijuana Legalization and Mental Health

Next, we review my estimates of RML adoption on my various mental health outcomes of interest. In Tables 3–6, I present 7 different regression specifications for each outcome. The first is my most basic specification, regressing the log outcome against a treatment variable for whether an RML is in effect along with state and time fixed effects. The next 6 specifications include one or more alterations to the first model: adding a control for whether an MML is in effect, restricting the sample of states to

only include MML-adopting states, including time-varying state-level covariates, and including region-specific time trends, with my last specification including all four of those additions.

Table 3. Log Proportion Reporting Any Mental Illness in the Past Year, Ages 18+

	All States		MML Adopters				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
RML in Effect	0.0486***	0.0481***	0.0463***	0.0427**	0.0461**	0.0474	0.0487*
	(0.0129)	(0.0139)	(0.0159)	(0.0193)	(0.0215)	(0.0281)	(0.0286)
MML in Effect		-0.00174		-0.00571		0.00200	0.00356
		(0.0147)		(0.0172)		(0.0194)	(0.0184)
State & Time FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State-Level Covariates	No	No	No	No	Yes	Yes	Yes
Region-Specific Time Trends	No	No	No	No	No	No	Yes
Sample Mean	0.1846	0.1846	0.1845	0.1845	0.1845	0.1845	0.1845
Observations	306	306	192	192	192	192	192
R-Squared	0.449	0.449	0.531	0.532	0.566	0.566	0.567

Notes: * p < 0.10, ** p < 0.05, *** p < 0.01.
All regressions include state and year fixed effects and are weighted by state population. Time-varying state-specific covariates include share of the population that is female, share that is white, share aged 18-64, share aged 65+, share with a college degree, unemployment rate, per capita income, alcohol tax rate, and share with health insurance. Standard errors, allowing for clustering at the state level, are reported in parentheses.

In Table 3, we observe that RML adoption is associated with increases in the log rate of any mental illness, with estimates ranging from 0.0427–0.0487, equivalent to a 4-5 percent increase. All the coefficient estimates for RMLs are statistically significant at the 10% level or lower, with the exception of specification (6).

I present my estimates of RML adoption on the rate of serious mental illness in Table 4. While RML adoption is associated with increases in the serious mental illness in the range of 2–4 percent, my estimates are only significant at the 10% level for the first two specifications. After restricting the sample to MML adopters and adding additional controls, these results are no longer significant.

Table 4. Log Proportion Reporting Serious Mental Illness in the Past Year, Ages 18+

	All States		MML Adopters				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
RML in Effect	0.0336*	0.0290*	0.0335	0.0202	0.0426	0.0270	0.0221
	(0.0179)	(0.0168)	(0.0216)	(0.0214)	(0.0318)	(0.0296)	(0.0307)
MML in Effect		-0.0144		-0.0212		-0.0236	-0.0298
		(0.0201)		(0.0249)		(0.0293)	(0.0283)
State & Time FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State-Level Covariates	No	No	No	No	Yes	Yes	Yes
Region-Specific Time Trends	No	No	No	No	No	No	Yes
Sample Mean	0.0419	0.0419	0.0413	0.0413	0.0413	0.0413	0.0413
Observations	306	306	192	192	192	192	192
R-Squared	0.691	0.692	0.716	0.717	0.738	0.740	0.743

Notes: * p < 0.10, ** p < 0.05, *** p < 0.01.
All regressions include state and year fixed effects and are weighted by state population. Time-varying state-specific covariates include share of the population that is female, share that is white, share aged 18-64, share aged 65+, share with a college degree, unemployment rate, per capita income, alcohol tax rate, and share with health insurance. Standard errors, allowing for clustering at the state level, are reported in parentheses.

Table 5 reports the effect of RML adoption on major depressive episodes. Here we see that RMLs increase the prevalence of major depressive episodes by 5–8 percent. All these estimates are significant at the 5% level or lower, with the exception of specification (4).

Table 5. Log Proportion Reporting a Major Depressive Episode in the Past Year, Ages 18+

	All States		MML Adopters				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
RML in Effect	0.0571***	0.0579***	0.0483**	0.0393	0.0803***	0.0752**	0.0783**
	(0.0154)	(0.0179)	(0.0182)	(0.0234)	(0.0250)	(0.0326)	(0.0325)
MML in Effect		0.00263		-0.0142		-0.00766	-0.00248
		(0.0192)		(0.0214)		(0.0242)	(0.0246)
State & Time FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State-Level Covariates	No	No	No	No	Yes	Yes	Yes
Region-Specific Time Trends	No	No	No	No	No	No	Yes
Sample Mean	0.0683	0.0683	0.0679	0.0679	0.0679	0.0679	0.0679
Observations	306	306	192	192	192	192	192
R-Squared	0.433	0.433	0.514	0.516	0.576	0.576	0.581

Notes: * p < 0.10, ** p < 0.05, *** p < 0.01.
All regressions include state and year fixed effects and are weighted by state population. Time-varying state-specific covariates include share of the population that is female, share that is white, share aged 18-64, share aged 65+, share with a college degree, unemployment rate, per capita income, alcohol tax rate, and share with health insurance. Standard errors, allowing for clustering at the state level, are reported in parentheses.

Finally, Table 6 shows the effect of RML adoption on suicidal ideation. Legalizing recreational marijuana was associated with increases in the rate of people experiencing serious thoughts of suicide of 5–9 percent, with all of these estimates being significant at the 10% level or lower.

Table 6. Log Proportion Reporting Serious Thoughts of Suicide in the Past Year, Ages 18+

	All States		MML Adopters				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
RML in Effect	0.0596***	0.0468**	0.0748***	0.0571*	0.0827**	0.0670*	0.0674*
	(0.0200)	(0.0180)	(0.0253)	(0.0296)	(0.0336)	(0.0355)	(0.0347)
MML in Effect		-0.0398**		-0.0281		-0.0237	-0.0249
		(0.0183)		(0.0256)		(0.0283)	(0.0274)
State & Time FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State-Level Covariates	No	No	No	No	Yes	Yes	Yes
Region-Specific Time Trends	No	No	No	No	No	No	Yes
Sample Mean	0.0402	0.0402	0.0403	0.0403	0.0403	0.0403	0.0403
Observations	306	306	192	192	192	192	192
R-Squared	0.588	0.595	0.620	0.624	0.639	0.641	0.649

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.
All regressions include state and year fixed effects and are weighted by state population. Time-varying state-specific covariates include share of the population that is female, share that is white, share aged 18-64, share aged 65+, share with a college degree, unemployment rate, per capita income, alcohol tax rate, and share with health insurance. Standard errors, allowing for clustering at the state level, are reported in parentheses.

In addition to the estimates presented in this section, I also run an expansive number of additional specifications for robustness. These include unweighted regressions, regressions using measures of mental health for different age groups (18–25 and 26+), regressions using binary indicators for the treatment variables rather than the fraction of the period, and regressions including an additional control for the fraction of the period that a state had active and legally protected medical dispensaries.

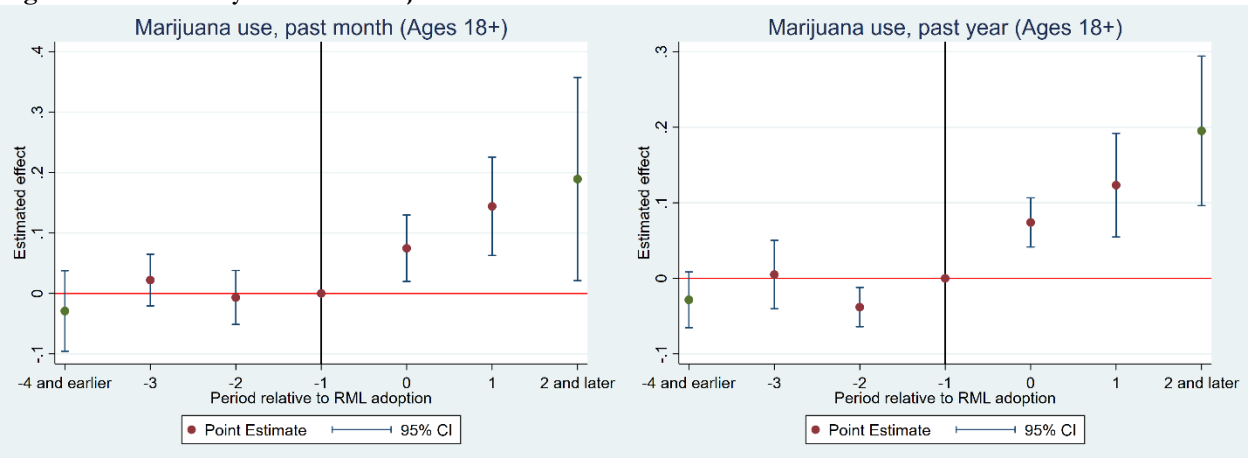
None of these models show qualitatively dissimilar results, with the exception of my regressions on 18–25-year-olds, which don't appear to have the pattern of significant declines in mental health outcomes associated with RML adoption that I find for adults 18+ and 26+. The most likely explanation for this is that young people already had relatively high rates of marijuana use prior to recreational legalization, so legalizing did not have as significant an effect on 18–25-year-olds as it did for the 26+ cohort, who seem to be driving the majority of my results. I present regression tables for 26+ and 18–25-year-olds in Appendix C.

4.3 Event Studies

In Figures 2 and 3, I present event study plots for my outcomes of interest which we can use to visualize the causal impact of RML adoption on marijuana use and mental health as well as identify any pre-treatment trends that might violate the parallel trends assumption. All of these plots were generated with the features of Specification (7): controlling for whether an MML is in effect, restricting the sample of states to only include MML-adopting states, including time-varying state-level covariates,

and including region-specific time trends. However, I also ran event studies using my other specifications and got substantively similar results.

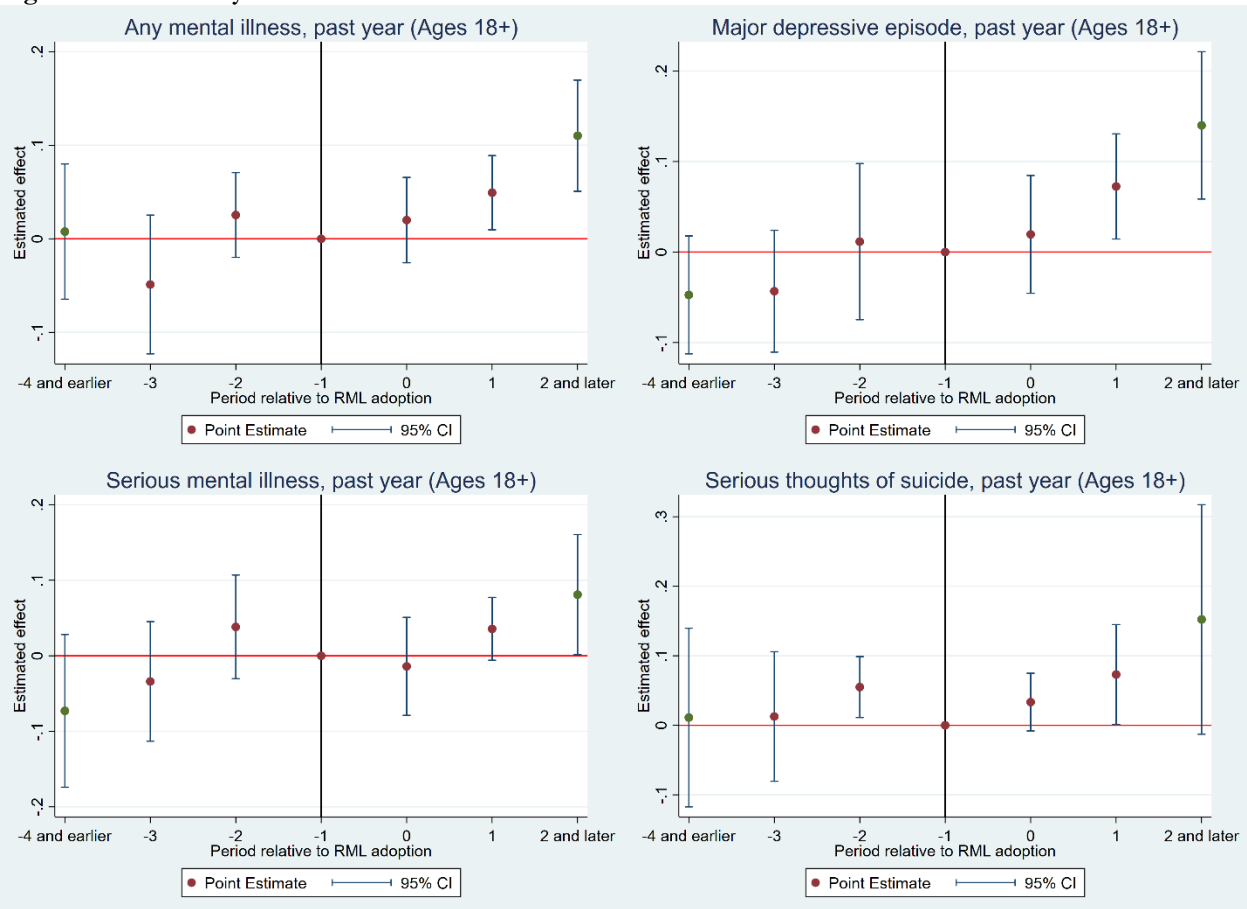
Figure 2. Event Study Plots for Marijuana Use



Notes: Event study regressions include state and year fixed effects, region time trends, and state-specific covariates; are restricted to RML adopters only; and are weighted by state population. Time-varying state-specific covariates include share of the population that is female, share that is white, share aged 18-64, share aged 65+, share with a college degree, unemployment rate, per capita income, alcohol tax rate, and share with health insurance. 95 percent confidence intervals are estimated using clustered standard errors.

Figure 2 shows how RML adoption led to immediate significant increases in marijuana use, though we see minor evidence of pre-trends in past year marijuana use. Figure 3 shows event study plots for my measures of mental health. One thing to note is that since each period consists of a two-year span of time, the effect estimated in period 0 is likely biased downward by states which had an RML go into effect towards the end of a 2-year period. But we see significantly higher rates of any mental illness, major depressive episodes, and serious thoughts of suicide by period 1. In general, the event study plots show patterns consistent with the results from Sections 4.1–4.2. They all also provide evidence in support of the parallel trends assumption, with some possible evidence of violation in the event studies for serious thoughts of suicide.

Figure 3. Event Study Plots for Mental Health Outcomes of Interest

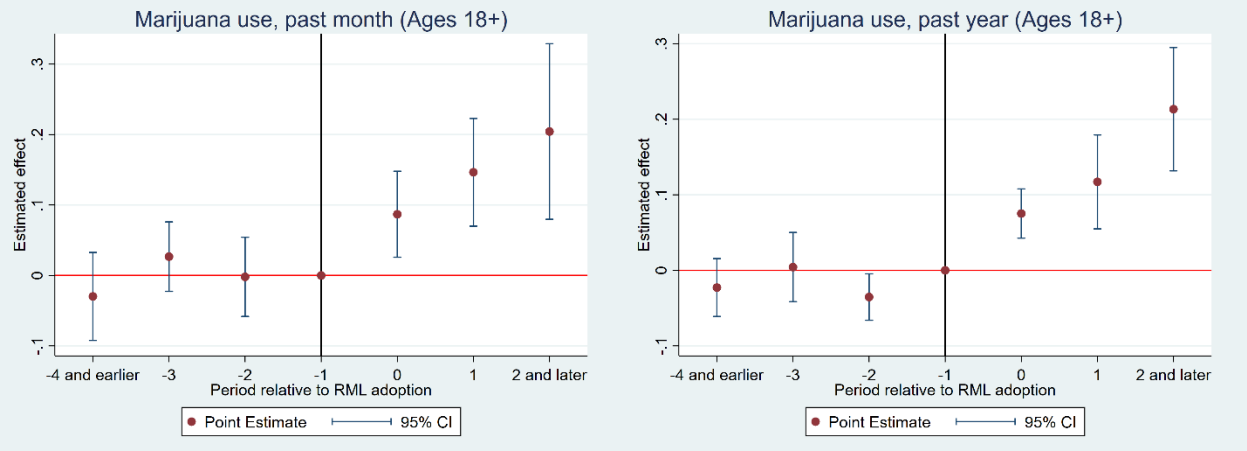


Notes: Event study regressions include state and year fixed effects, region time trends, and state-specific covariates; are restricted to RML adopters only; and are weighted by state population. Time-varying state-specific covariates include share of the population that is female, share that is white, share aged 18-64, share aged 65+, share with a college degree, unemployment rate, per capita income, alcohol tax rate, and share with health insurance. 95 percent confidence intervals are estimated using clustered standard errors.

4.4 Robustness

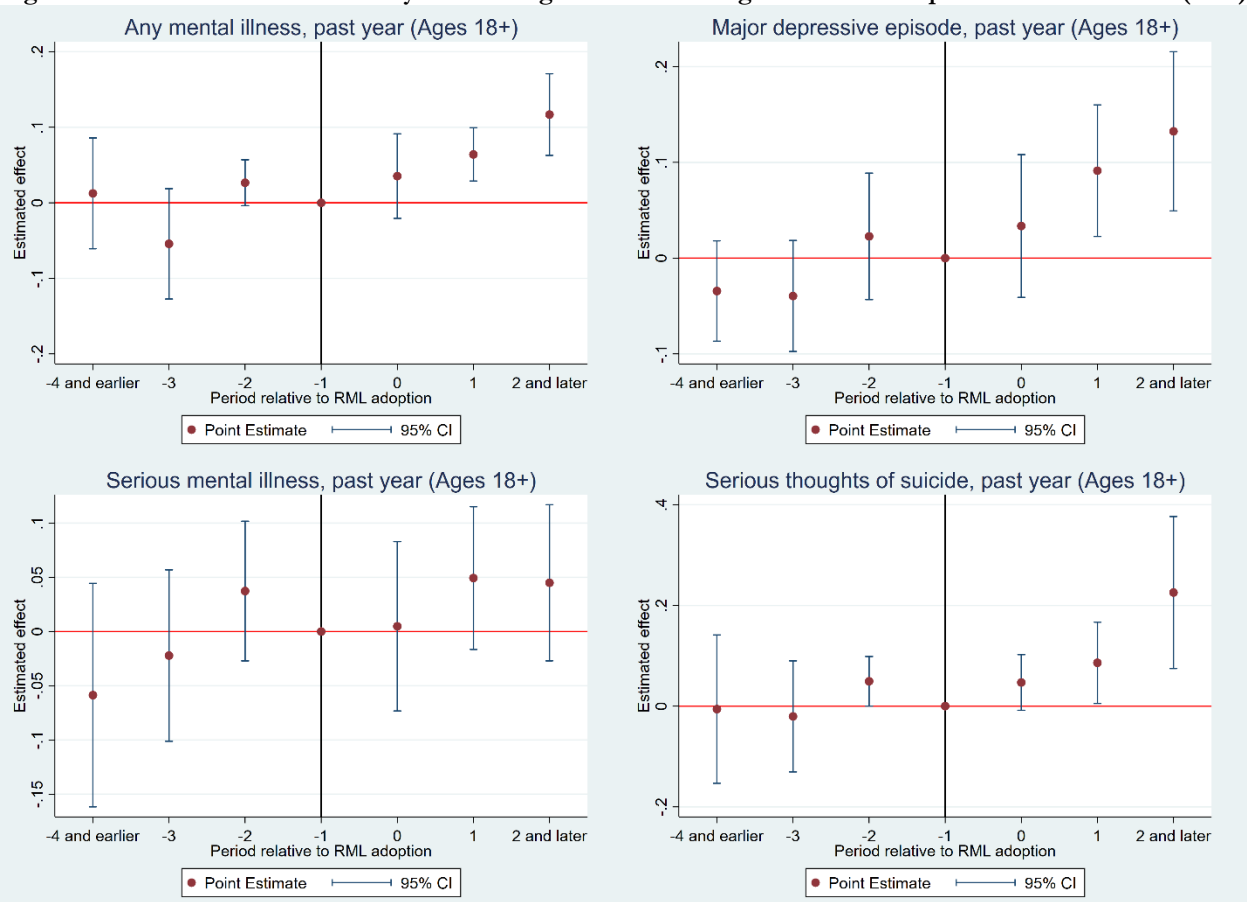
Recent work has highlighted that the two-way fixed effects estimator commonly used for staggered difference-in-differences estimation relies on the strong assumption that treatment effects are constant over time in order to give it a causal interpretation (Goodman-Bacon, 2021). Advances in the econometrics literature have developed estimators that address the problems of two-way fixed effects regressions in the presence of heterogenous treatment effects. In Figures 4 and 5, I present event study plots for my main outcomes of interest using the interaction-weighted estimator proposed by Sun & Abraham (2021). Given their qualitative similarity to the event study plots in the previous section, it's unlikely that my results are being driven by treatment effects heterogeneity.

Figure 4. Marijuana Use Event Study Plots Using Interaction-Weighted Estimator per Sun and Abraham (2021)



Notes: Event study regressions include state and year fixed effects, region time trends, and state-specific covariates; are restricted to RML adopters only; and are weighted by state population. Time-varying state-specific covariates include share of the population that is female, share that is white, share aged 18-64, share aged 65+, share with a college degree, unemployment rate, per capita income, alcohol tax rate, and share with health insurance. 95 percent confidence intervals are estimated using clustered standard errors.

Figure 5. Mental Health Event Study Plots Using Interaction-Weighted Estimator per Sun and Abraham (2021)



Notes: Event study regressions include state and year fixed effects, region time trends, and state-specific covariates; are restricted to RML adopters only; and are weighted by state population. Time-varying state-specific covariates include share of the population that is female, share that is white, share aged 18-64, share aged 65+, share with a college degree, unemployment rate, per capita income, alcohol tax rate, and share with health insurance. 95 percent confidence intervals are estimated using clustered standard errors.

5. Discussion

The results I describe stand in stark contrast to existing research in this area, mainly focused on MMLs, which find that MMLs led to improvements (Anderson, Rees, & Sabia, 2014; Bartos, Kubrin, Newark, & McCleary, 2020; Kalbfuß, Odermatt, & Stutzer, 2018) or had no impact (Grucza et al., 2015; Leung, 2019) on mental health outcomes. Based on the results of those studies, the evidence I present linking RMLs to declines in mental health come as a surprise. Unfortunately, due to limitations in the data available to me, I am unable to identify the specific mechanism driving these apparently contradictory results and can only propose some plausible explanations.

The most convincing explanation is that since medical and recreational laws target different groups by design, their differing effects on mental health could reflect heterogeneous effects of marijuana use on different subpopulations. MMLs expand access to people with medical conditions such as chronic pain or epilepsy for which there is at least some belief in the efficacy of marijuana as a treatment. For these individuals, any potential side effects from marijuana use would likely be outweighed by substantial improvements to their quality of life. Additionally, since MMLs require prospective patients to interact with a physician at least once before getting to purchase marijuana, it is possible that the presence of a medical professional reduces the risk of unhealthy marijuana use. On the other hand, RMLs expand access to the entire adult population, and individuals can purchase and use marijuana independent of whether it's likely to confer a health benefit.

Another possible explanation is that the presence of RMLs affect behavior other than marijuana use that negatively impacts mental health. For instance, increased access to marijuana might lead people to neglect treatment for mental health disorders, preferring to self-medicate with marijuana instead. However, I find no evidence that RMLs are associated with changes in the share of the population receiving mental health services. Table 7 presents regressions estimating the effect of RML adoption on the log share of adults receiving mental health services. I find that the coefficient for RML in effect is neither significant nor has a consistent sign across any of the specifications estimated. I present event study plots in Appendix B confirming these results.

Table 7. Log Proportion Receiving Mental Health Services in the Past Year, Ages 18+

	All States		MML Adopters				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
RML in Effect	0.0110	0.0243	-0.00647	0.0121	-0.0204	-0.00993	-0.00668
	(0.0195)	(0.0200)	(0.0212)	(0.0271)	(0.0269)	(0.0320)	(0.0308)
MML in Effect		0.0405**		0.0293		0.0149	0.0223
		(0.0162)		(0.0173)		(0.0186)	(0.0207)
State & Time FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State-Level Covariates	No	No	No	No	Yes	Yes	Yes
Region-Specific Time Trends	No	No	No	No	No	No	Yes
Sample Mean	0.1459	0.1459	0.1455	0.1455	0.1455	0.1455	0.1455
Observations	255	255	160	160	160	160	160
R-Squared	0.463	0.483	0.518	0.527	0.560	0.562	0.589

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.
All regressions include state and year fixed effects and are weighted by state population. Time-varying state-specific covariates include share of the population that is female, share that is white, share aged 18-64, share aged 65+, share with a college degree, unemployment rate, per capita income, alcohol tax rate, and share with health insurance. Standard errors, allowing for clustering at the state level, are reported in parentheses.

It is also possible that my results are driven by biases due to limitations in my research design. There is, for example, evidence of spillover effects from marijuana legalization which may affect the validity of my estimates. Hao & Cowan (2019) find that following recreational legalization in Colorado and Washington, marijuana use increased in neighboring states relative to non-neighboring states. They also find that arrests for marijuana possession increased sharply in those neighboring states' border counties relative to non-bordering ones. Hansen, Miller, & Weber (2018) look at recreational marijuana sales by Washington dispensaries along the Oregon border and identify large declines in sales immediately after Oregon opened its own recreational dispensaries, implying that marijuana had previously been trafficked from recreationally legal Washington into recreationally illegal Oregon. While these potential spillovers might complicate estimation of an accurate effect from RMLs, we would expect their presence to bias my estimates towards zero, and so the true effect may be even larger than presented.

Although my results are robust to the inclusion of demographic controls and other state-specific covariates, I cannot rule out the possibility that some of my results are driven by unobserved changes in the composition of state populations. One particularly interesting consideration is whether there is any effect from migration spurred by legalization. Zambiasi & Stillman (2020) find that after the 2009 Ogden memo allowed Colorado's marijuana laws to take effect, the state experienced higher in-migration than would otherwise be expected. It is possible that other states experienced similar effects after legalization. However, this is only relevant if we believe that legalization-driven migration would attract people with a higher rate of mental health disorders controlling for my existing covariates.

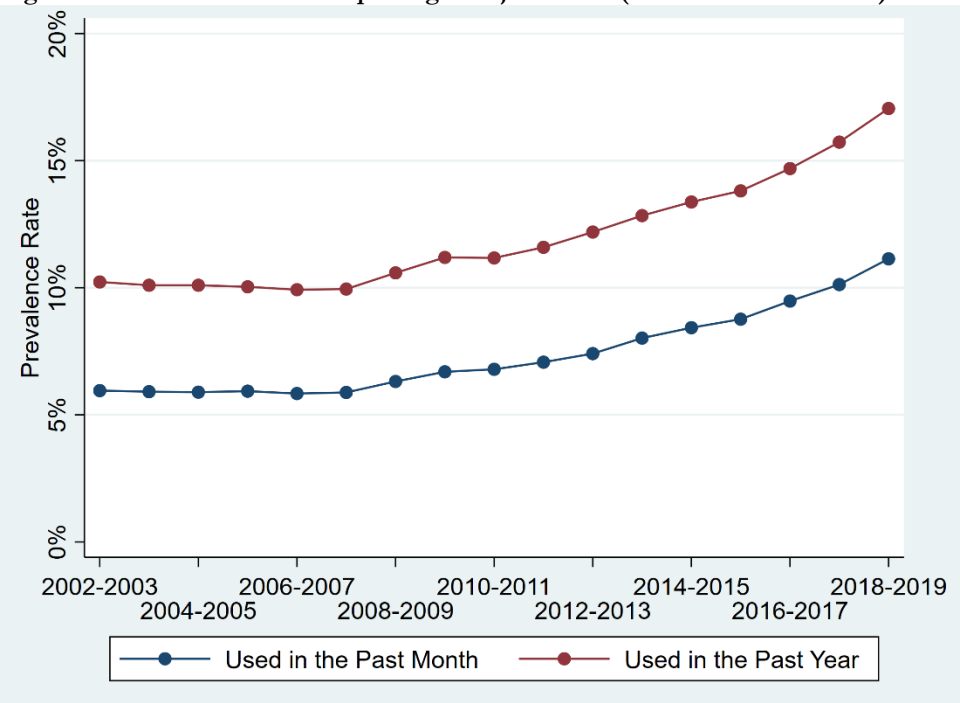
Additionally, Zambiasi & Stillman identify increased migration into Colorado in 2009–2012, during which time the state only had legal medical marijuana. So, were the migration explanation true, we would expect MMLs to also lead to declines in mental health, which is contrary to the evidence.

Even if we rule out alternative explanations driving spurious results, there are still many limitations stemming from the measures of mental health which I rely upon. For one, while my results give some indication as to changes in prevalence of mental illness in the population, they give only limited information about what specific disorders are associated with RMLs and their severity. This same issue of intensive margin identification extends to my measures of marijuana use. While I observe increases in the share of people who consume marijuana following RML adoption, I do not observe how much they consume. It would not be surprising, for instance, if declines in mental health due to RMLs were primarily driven by increases in marijuana consumption among the most prolific users. Additionally, given that I identify possible heterogeneity in the effect of marijuana legalization on medical users versus recreational users, it is likely that the population of recreational users themselves exhibit different effects across subpopulations. Individual-level data, such as the NSDUH restricted-use data, could help shed light on questions of whether certain behaviors, such as frequency of use, or specific identifiable characteristics are risk factors for the harms I identify. Future research might use that as a starting point for clarifying the relationship between RMLs and mental health.

But while these limitations make it difficult to be conclusive about how precisely RMLs affect mental health, the consistency of the evidence across different measures of mental health, the robustness of my results across different specifications, and the lack of convincing alternative explanations lends likelihood to the possibility that marijuana use is harmful to mental health. This is not to say that RMLs are harmful overall. The expanding literature on marijuana policy has found multiple examples of public health benefits associated with legalization, including reductions in alcohol consumption (Anderson, Hansen, & Rees, 2013; Miller & Seo, 2021), traffic fatalities (Anderson, Hansen, & Rees, 2013; Hansen, Miller, & Weber, 2020), tobacco consumption (Choi, Dave, & Sabia, 2019; Miller & Seo, 2021) and opioid mortalities (Bachhuber et al., 2014; Powell, Pacula, & Jacobson, 2018), not to mention its possible benefits in other spheres of American life. Whether these benefits outweigh the costs is a normative question outside the purview of this paper. But policymakers should use additional caution when considering implementation of RMLs and should also be prepared for challenges that may accompany the rising use of marijuana in the US (See Figure 6). Further research examining the mechanisms by which RMLs impact mental health and identifying characteristics which put

individuals at greater risk of marijuana's mental health harms may inform policy to mitigate recreational legalization's negative effects.

Figure 6. Share of US Adults Reporting Marijuana Use (2002-2003 to 2018-2019)



Notes: Data from the National Survey of Drug Use and Health (2002-2019).

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Appendix

Appendix A. Table of Medical and Recreational Marijuana Laws by State

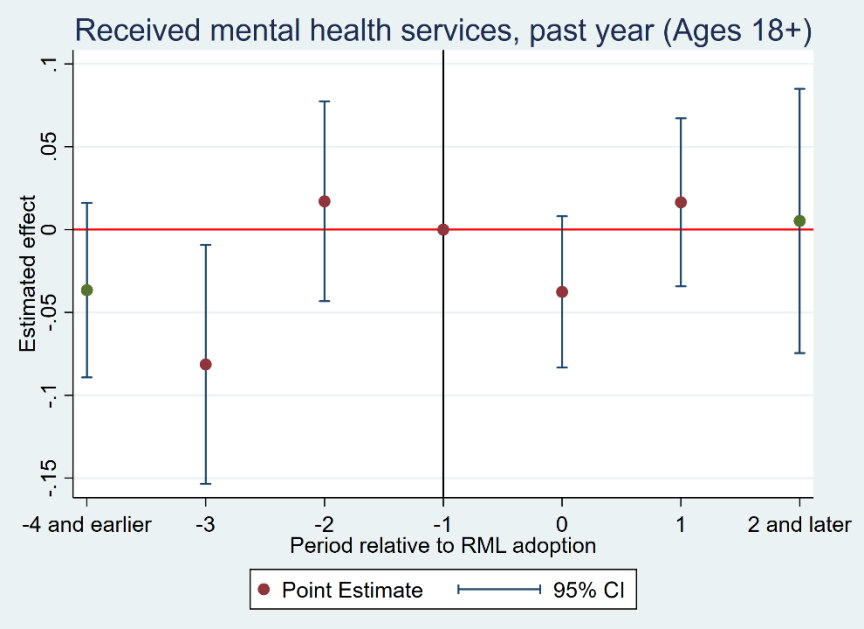
Table A1. Medical and Recreational Marijuana Laws by State, 1996 to 2021

	MML Effective Date	RML Effective Date	Recreational Sales Allowed
Alaska	3/4/1999	2/24/2015	10/29/2016
Arizona	4/14/2011	11/30/2020	
Arkansas	11/9/2016		
California	11/6/1996	11/9/2016	1/1/2018
Colorado	6/1/2001	12/10/2012	1/1/2014
Connecticut	8/20/2014		
Delaware	6/26/2015		
District of Columbia	7/30/2013	2/26/2015	2/26/2015
Florida	7/26/2016		
Hawaii	12/28/2000		
Illinois	11/9/2015	1/1/2020	1/1/2020
Louisiana	8/6/2019		
Maine	12/22/1999	1/31/2017	
Maryland	12/2/2017		
Massachusetts	1/1/2013	12/15/2016	11/20/2018
Michigan	12/4/2008	12/6/2018	12/1/2019
Minnesota	7/1/2015		
Mississippi	Not yet operational		
Missouri	10/17/2020		
Montana	11/2/2004	1/1/2021	
Nevada	10/1/2001	1/1/2017	7/1/2017
New Hampshire	5/1/2016		
New Jersey	12/6/2012	2/22/2021	
New Mexico	7/1/2007		
New York	1/8/2016		
North Dakota	3/1/2019		
Ohio	1/16/2019		
Oklahoma	7/26/2018		
Oregon	12/3/1998	7/1/2015	10/1/2015
Pennsylvania	1/17/2018		
Rhode Island	1/3/2006		
South Dakota	Not yet operational		
Utah	3/2/2020		
Vermont	7/1/2004	7/1/2018	
Virginia	10/17/2020		
Washington	11/3/1998	12/6/2012	7/8/2014
West Virginia	8/22/2017		

Source: Anderson & Rees (2021)

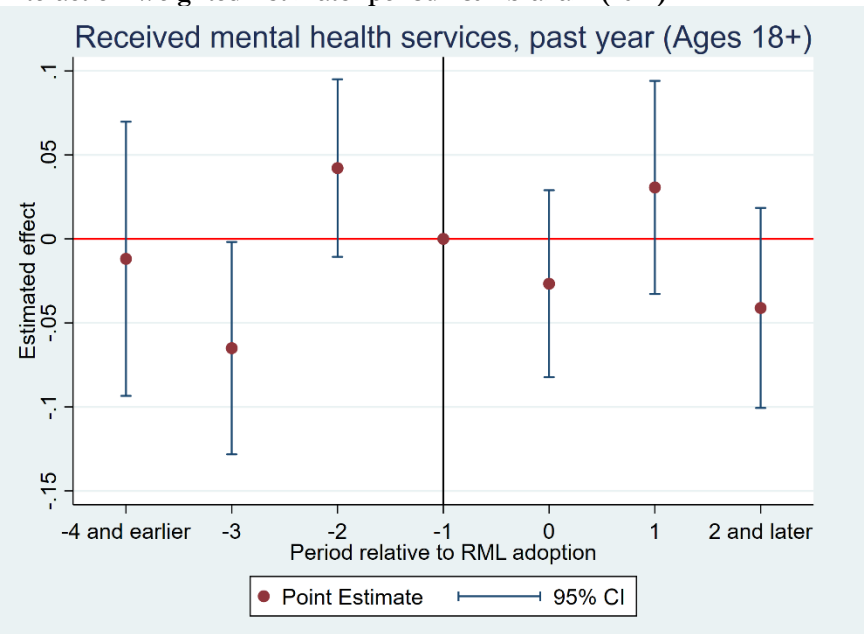
Appendix B. Event Study Plots for Mental Health Services Utilization

Figure A1. Event Study Plot for Mental Health Services Utilization



Notes: Event study regressions include state and year fixed effects, region time trends, and state-specific covariates; are restricted to RML adopters only; and are weighted by state population. Time-varying state-specific covariates include share of the population that is female, share that is white, share aged 18-64, share aged 65+, share with a college degree, unemployment rate, per capita income, alcohol tax rate, and share with health insurance. 95 percent confidence intervals are estimated using clustered standard errors.

Figure A2. Event Study Plot of Mental Health Services Utilization Using Interaction-Weighted Estimator per Sun & Abraham (2021)



Notes: Event study regressions include state and year fixed effects, region time trends, and state-specific covariates; are restricted to RML adopters only; and are weighted by state population. Time-varying state-specific covariates include share of the population that is female, share that is white, share aged 18-64, share aged 65+, share with a college degree, unemployment rate, per capita income, alcohol tax rate, and share with health insurance. 95 percent confidence intervals are estimated using clustered standard errors.

Appendix C. Regressions on Mental Health Outcomes for Other Age Groups

Table A2. Log Proportion Reporting Any Mental Illness in the Past Year, Ages 26+

	All States		MML Adopters				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
RML in Effect	0.0594***	0.0582***	0.0558***	0.0486**	0.0537**	0.0530	0.0537
	(0.0133)	(0.0152)	(0.0159)	(0.0213)	(0.0226)	(0.0319)	(0.0323)
MML in Effect		-0.00379		-0.0115		-0.00110	-0.000295
		(0.0154)		(0.0192)		(0.0225)	(0.0213)
State & Time FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State-Level Covariates	No	No	No	No	Yes	Yes	Yes
Region-Specific Time Trends	No	No	No	No	No	No	Yes
Sample Mean	0.1797	0.1797	0.1791	0.1791	0.1791	0.1791	0.1791
Observations	306	306	192	192	192	192	192
R-Squared	0.210	0.210	0.305	0.307	0.355	0.355	0.356

Notes: * p < 0.10, ** p < 0.05, *** p < 0.01.
All regressions include state and year fixed effects and are weighted by state population. Time-varying state-specific covariates include share of the population that is female, share that is white, share aged 18-64, share aged 65+, share with a college degree, unemployment rate, per capita income, alcohol tax rate, and share with health insurance. Standard errors, allowing for clustering at the state level, are reported in parentheses.

Table A3. Log Proportion Reporting Serious Mental Illness in the Past Year, Ages 26+

	All States		MML Adopters				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
RML in Effect	0.0402	0.0330	0.0393	0.0163	0.0437	0.0186	0.0133
	(0.0241)	(0.0229)	(0.0263)	(0.0251)	(0.0334)	(0.0326)	(0.0338)
MML in Effect		-0.0224		-0.0364		-0.0380	-0.0449
		(0.0197)		(0.0256)		(0.0326)	(0.0320)
State & Time FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State-Level Covariates	No	No	No	No	Yes	Yes	Yes
Region-Specific Time Trends	No	No	No	No	No	No	Yes
Sample Mean	0.0402	0.0402	0.0395	0.0395	0.0395	0.0395	0.0395
Observations	306	306	192	192	192	192	192
R-Squared	0.381	0.384	0.410	0.418	0.448	0.455	0.462

Notes: * p < 0.10, ** p < 0.05, *** p < 0.01.
All regressions include state and year fixed effects and are weighted by state population. Time-varying state-specific covariates include share of the population that is female, share that is white, share aged 18-64, share aged 65+, share with a college degree, unemployment rate, per capita income, alcohol tax rate, and share with health insurance. Standard errors, allowing for clustering at the state level, are reported in parentheses.

Table A4. Log Proportion Reporting a Major Depressive Episode in the Past Year, Ages 26+

	All States		MML Adopters				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
RML in Effect	0.0735***	0.0728***	0.0628***	0.0465	0.101***	0.0914**	0.0945**
	(0.0206)	(0.0247)	(0.0219)	(0.0295)	(0.0311)	(0.0413)	(0.0413)
MML in Effect		-0.00220		-0.0259		-0.0141	-0.00888
		(0.0213)		(0.0239)		(0.0286)	(0.0291)
State & Time FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State-Level Covariates	No	No	No	No	Yes	Yes	Yes
Region-Specific Time Trends	No	No	No	No	No	No	Yes
Sample Mean	0.0627	0.0627	0.062	0.062	0.062	0.062	0.062
Observations	306	306	192	192	192	192	192
R-Squared	0.159	0.159	0.175	0.182	0.259	0.260	0.266

Notes: * p < 0.10, ** p < 0.05, *** p < 0.01.
All regressions include state and year fixed effects and are weighted by state population. Time-varying state-specific covariates include share of the population that is female, share that is white, share aged 18-64, share aged 65+, share with a college degree, unemployment rate, per capita income, alcohol tax rate, and share with health insurance. Standard errors, allowing for clustering at the state level, are reported in parentheses.

Table A5. Log Proportion Reporting Serious Thoughts of Suicide in the Past Year, Ages 26+

	All States		MML Adopters				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
RML in Effect	0.104***	0.0850***	0.116***	0.0756*	0.106***	0.0669*	0.0672
	(0.0341)	(0.0300)	(0.0375)	(0.0383)	(0.0335)	(0.0390)	(0.0401)
MML in Effect		-0.0585***		-0.0648*		-0.0594	-0.0618
		(0.0199)		(0.0336)		(0.0367)	(0.0371)
State & Time FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State-Level Covariates	No	No	No	No	Yes	Yes	Yes
Region-Specific Time Trends	No	No	No	No	No	No	Yes
Sample Mean	0.0332	0.0332	0.0333	0.0333	0.0333	0.0333	0.0333
Observations	306	306	192	192	192	192	192
R-Squared	0.158	0.180	0.244	0.269	0.276	0.293	0.312

Notes: * p < 0.10, ** p < 0.05, *** p < 0.01.
All regressions include state and year fixed effects and are weighted by state population. Time-varying state-specific covariates include share of the population that is female, share that is white, share aged 18-64, share aged 65+, share with a college degree, unemployment rate, per capita income, alcohol tax rate, and share with health insurance. Standard errors, allowing for clustering at the state level, are reported in parentheses.

Table A6. Log Proportion Receiving Mental Health Services in the Past Year, Ages 26+

	All States		MML Adopters				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
RML in Effect	0.00904	0.0234	-0.0106	0.00817	-0.0252	-0.0176	-0.0141
	(0.0215)	(0.0220)	(0.0235)	(0.0300)	(0.0303)	(0.0363)	(0.0345)
MML in Effect		0.0436**		0.0296		0.0108	0.0186
		(0.0180)		(0.0191)		(0.0197)	(0.0222)
State & Time FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State-Level Covariates	No	No	No	No	Yes	Yes	Yes
Region-Specific Time Trends	No	No	No	No	No	No	Yes
Sample Mean	0.1485	0.1485	0.1479	0.1479	0.1479	0.1479	0.1479
Observations	255	255	160	160	160	160	160
R-Squared	0.289	0.314	0.330	0.340	0.395	0.396	0.433

Notes: * p < 0.10, ** p < 0.05, *** p < 0.01.
All regressions include state and year fixed effects and are weighted by state population. Time-varying state-specific covariates include share of the population that is female, share that is white, share aged 18-64, share aged 65+, share with a college degree, unemployment rate, per capita income, alcohol tax rate, and share with health insurance. Standard errors, allowing for clustering at the state level, are reported in parentheses.

Table A7. Log Proportion Reporting Any Mental Illness in the Past Year, Ages 18-25

	All States		MML Adopters				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
RML in Effect	-0.00559	-0.000931	-0.00673	0.00715	0.00490	0.0171	0.0195
	(0.0270)	(0.0260)	(0.0281)	(0.0267)	(0.0289)	(0.0294)	(0.0309)
MML in Effect		0.0145		0.0220		0.0184	0.0216
		(0.0171)		(0.0196)		(0.0230)	(0.0233)
State & Time FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State-Level Covariates	No	No	No	No	Yes	Yes	Yes
Region-Specific Time Trends	No	No	No	No	No	No	Yes
Sample Mean	0.2145	0.2145	0.2183	0.2183	0.2183	0.2183	0.2183
Observations	306	306	192	192	192	192	192
R-Squared	0.922	0.923	0.922	0.922	0.926	0.926	0.927

Notes: * p < 0.10, ** p < 0.05, *** p < 0.01.
All regressions include state and year fixed effects and are weighted by state population. Time-varying state-specific covariates include share of the population that is female, share that is white, share aged 18-64, share aged 65+, share with a college degree, unemployment rate, per capita income, alcohol tax rate, and share with health insurance. Standard errors, allowing for clustering at the state level, are reported in parentheses.

Table A8. Log Proportion Reporting Serious Mental Illness in the Past Year, Ages 18-25

	All States		MML Adopters				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
RML in Effect	0.00467	0.0107	0.00853	0.0357	0.0544	0.0850**	0.0826**
	(0.0402)	(0.0382)	(0.0426)	(0.0411)	(0.0405)	(0.0365)	(0.0366)
MML in Effect		0.0187		0.0431		0.0461	0.0420
		(0.0268)		(0.0358)		(0.0308)	(0.0283)
State & Time FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State-Level Covariates	No	No	No	No	Yes	Yes	Yes
Region-Specific Time Trends	No	No	No	No	No	No	Yes
Sample Mean	0.0522	0.0522	0.0529	0.0529	0.0529	0.0529	0.0529
Observations	306	306	192	192	192	192	192
R-Squared	0.962	0.962	0.962	0.963	0.968	0.968	0.969

Notes: * p < 0.10, ** p < 0.05, *** p < 0.01.
All regressions include state and year fixed effects and are weighted by state population. Time-varying state-specific covariates include share of the population that is female, share that is white, share aged 18-64, share aged 65+, share with a college degree, unemployment rate, per capita income, alcohol tax rate, and share with health insurance. Standard errors, allowing for clustering at the state level, are reported in parentheses.

Table A9. Log Proportion Reporting a Major Depressive Episode in the Past Year, Ages 18-25

	All States		MML Adopters				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
RML in Effect	-0.0000969	0.00903	-0.00941	0.00776	0.00983	0.0211	0.0231
	(0.0332)	(0.0319)	(0.0347)	(0.0376)	(0.0451)	(0.0483)	(0.0494)
MML in Effect		0.0285		0.0273		0.0171	0.0194
		(0.0275)		(0.0380)		(0.0387)	(0.0385)
State & Time FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State-Level Covariates	No	No	No	No	Yes	Yes	Yes
Region-Specific Time Trends	No	No	No	No	No	No	Yes
Sample Mean	0.1026	0.1026	0.1043	0.1043	0.1043	0.1043	0.1043
Observations	306	306	192	192	192	192	192
R-Squared	0.924	0.925	0.923	0.924	0.931	0.932	0.932

Notes: * p < 0.10, ** p < 0.05, *** p < 0.01.
All regressions include state and year fixed effects and are weighted by state population. Time-varying state-specific covariates include share of the population that is female, share that is white, share aged 18-64, share aged 65+, share with a college degree, unemployment rate, per capita income, alcohol tax rate, and share with health insurance. Standard errors, allowing for clustering at the state level, are reported in parentheses.

Table A10. Log Proportion Reporting Serious Thoughts of Suicide in the Past Year, Ages 18-25

	All States		MML Adopters				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
RML in Effect	-0.0183	-0.0142	-0.00661	0.0253	0.0390	0.0763	0.0759
	(0.0490)	(0.0474)	(0.0509)	(0.0460)	(0.0482)	(0.0495)	(0.0504)
MML in Effect		0.0127		0.0507**		0.0563**	0.0553**
		(0.0239)		(0.0233)		(0.0257)	(0.0240)
State & Time FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State-Level Covariates	No	No	No	No	Yes	Yes	Yes
Region-Specific Time Trends	No	No	No	No	No	No	Yes
Sample Mean	0.0823	0.0823	0.0837	0.0837	0.0837	0.0837	0.0837
Observations	306	306	192	192	192	192	192
R-Squared	0.921	0.921	0.925	0.928	0.933	0.935	0.935

Notes: * p < 0.10, ** p < 0.05, *** p < 0.01.
All regressions include state and year fixed effects and are weighted by state population. Time-varying state-specific covariates include share of the population that is female, share that is white, share aged 18-64, share aged 65+, share with a college degree, unemployment rate, per capita income, alcohol tax rate, and share with health insurance. Standard errors, allowing for clustering at the state level, are reported in parentheses.

Table A11. Log Proportion Receiving Mental Health Services in the Past Year, Ages 18-25

	All States		MML Adopters				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
RML in Effect	0.0189	0.0282	0.0108	0.0295	0.00596	0.0381	0.0393
	(0.0184)	(0.0181)	(0.0228)	(0.0225)	(0.0320)	(0.0312)	(0.0295)
MML in Effect		0.0283		0.0294		0.0457	0.0498
		(0.0263)		(0.0283)		(0.0345)	(0.0326)
State & Time FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State-Level Covariates	No	No	No	No	Yes	Yes	Yes
Region-Specific Time Trends	No	No	No	No	No	No	Yes
Sample Mean	0.1303	0.1303	0.1316	0.1316	0.1316	0.1316	0.1316
Observations	255	255	160	160	160	160	160
R-Squared	0.855	0.856	0.873	0.874	0.877	0.880	0.881

Notes: * p < 0.10, ** p < 0.05, *** p < 0.01.
All regressions include state and year fixed effects and are weighted by state population. Time-varying state-specific covariates include share of the population that is female, share that is white, share aged 18-64, share aged 65+, share with a college degree, unemployment rate, per capita income, alcohol tax rate, and share with health insurance. Standard errors, allowing for clustering at the state level, are reported in parentheses.